Dr. Shafayat Abrar
Probability & Statistics

Dated: Jan. 11, 2024
.

The three axioms of probability

Axiom 1

$$0 \leq P(E) \leq 1$$

Axiom 2

$$P(\Omega) = 1$$

Axiom 3

For any sequence of mutually exclusive events $E_1, E_2, \ldots$
(that is, events for which $E_i \cap E_j = \varnothing$ when $i \neq j$),

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P\left(E_i\right)$$

We refer to $P(E)$ as the probability of the event E.

Proposition: If $A \subset B$, then $P(A) \leq P(B)$.

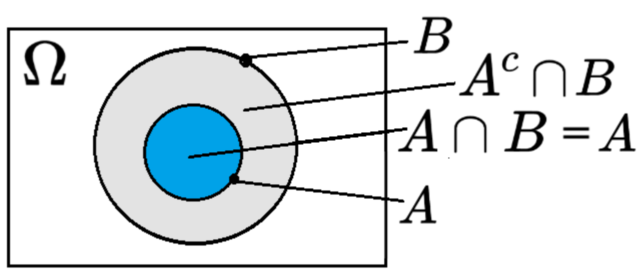
Proof: Since $A \subset B$, A is a proper subset of B, all elements of A are contained by B but the converse is not necessarily true; it follows that we can express B as

$$B = (A \cap B) \cup (A^c \cap B) = A \cup (A^c \cap B)$$

Because $A \cap B$ and $A^c \cap B$ are mutually exclusive, where $A^c \cap B$ is the region (or elements) that belongs to B but is outside A.; therefore, we obtain, from Axiom 3,

$$P(B) = P(A) + P\left(A^c \cap B\right) \geq P(A)$$

which proves the result, since $P\left(A^c \cap B\right) \geq 0$.



Proposition:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: Let us divide $A \cup B$ into three mutually exclusive sections, as shown in Figure below. In other words, section I represents all the points in A that are not in B (that is, $\mathrm{I} = A \cap B^c$), section II represents all points both in A and in B (that is, $\mathrm{II} = A \cap B$), and section III represents all points in B that are not in A (that is, $\mathrm{III} = A^c \cap B$). From Figure, we see that

$$A \cup B = \mathrm{I} \cup \mathrm{II} \cup \mathrm{III}$$
$$A = \mathrm{I} \cup \mathrm{II}$$
$$B = \mathrm{II} \cup \mathrm{III}$$

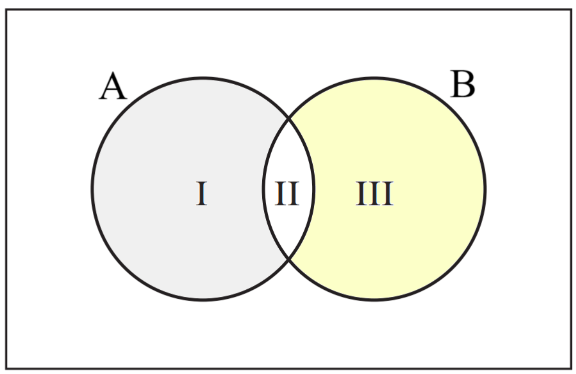
As I, II, and III are mutually exclusive, it follows from Axiom 3 that

$$P(A \cup B) = P(\mathrm{I}) + P(\mathrm{II}) + P(\mathrm{III})$$
$$P(A) = P(\mathrm{I}) + P(\mathrm{II})$$
$$P(B) = P(\mathrm{II}) + P(\mathrm{III})$$

which shows that

$$P(A \cup B) = P(A) + P(B) - P(\mathrm{II})$$

and the above Proposition is proved, since region $\mathrm{II} = A \cap B$.



Defining Events in terms of Set Notations:

Let A and B be two events. Find expressions for:

(a) Both A and B occur.

$$A \cap B$$

(b) None of the two events occur.

$$A^c \cap B^c = (A \cup B)^c$$

(c) At least one of the two events occurs.

$$A \cup B$$

(d) Exactly one of the two events occurs.

$$(A^c \cap B) \cup (A \cap B^c)$$

Let A, B, and C be three events. Find expressions for:

(a) Both A and B but not C occur.

$$A \cap B \cap C^c$$

or using simpler notations, one may also write

$$ABC^c$$

(b) None of the three events occur.

$$A^c \cap B^c \cap C^c = (A \cup B \cup C)^c$$

or using simpler notations, one may also write

$$A^c B^c C^c = (A \cup B \cup C)^c$$

(c) At least one of the three events occurs.

$$A \cup B \cup C$$

(d) Exactly one of the three events occurs.

$$(A^c \cap B \cap C^c) \cup (A \cap B^c \cap C^c) \cup (A^c \cap B^c \cap C)$$

or using simpler notations, one may also write

$$(A^c B C^c) \cup (A B^c C^c) \cup (A^c B^c C)$$